

Core Architectural Shift

Our biggest change is moving away from the past rendering system into one that uses realtime data playback more.

Major Changes

1. From Two Modes to One with UI Control

Current Design:

- Two modes: Collision-only and Collision-to-Doppler
- Command-line driven with YAML/JSON specifications
- Headless execution with saved bundles

New Vision:

- Single unified mode controlled via UI
- Real-time option available ONLY if the user setup an artifact source, else point to the setup option
- Direct mode selection from UI

2. Replacing Batch Scheduler with Video Player Metaphor

In the new design, we will move to something where we

- Add a timeline scrubber similar to YouTube's player
- Implement realtime playback at different speeds
- Auto clipping system to segment recordings into intervals (which can generate videos with TTS, this is pretty easy to do but for later)
- Visual playback controls (play/pause, seek, speed control)
- Chunked visual input, allowing you to choose specific frame chunks

3. New Storage Architecture

- Replace bundle-based storage with chunked recording system
- Add B-tree indexing for fast lookup of arbitrary timestamps
- Implement memory-mapped file access for efficient playback
- Add highlight/presentation system

New Components Needed

1. Real-time Data Consumer
2. Playback Controller
3. Storage Manager with B-tree Index
4. Auto-clipping System
5. (Extra) Highlight and Presentation System

Overall Goals

- Real-time 3D spatial data visualization
- YouTube-like playback experience
- Efficient timestamp-based lookup
- Auto-generated clips and presentations
- Memory-mapped file access for performance